EDS 223 Homework 2 Exploring patterns of environmental justice

2026-10-21

Preliminary Steps

Load the packages we will need for analysis.

```
# Load necessary packages
library(tidyverse)
```

```
Warning: package 'ggplot2' was built under R version 4.4.3

Warning: package 'tidyr' was built under R version 4.4.3

Warning: package 'purrr' was built under R version 4.4.3

Warning: package 'dplyr' was built under R version 4.4.3

Warning: package 'lubridate' was built under R version 4.4.3

-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.0     v readr     2.1.5
v forcats   1.0.0     v stringr   1.5.1
v ggplot2   4.0.2     v tibble    3.3.0
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.1
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to beco
```

```
library(dplyr)
library(sf)
```

```
Warning: package 'sf' was built under R version 4.4.3
```

```
Linking to GEOS 3.13.0, GDAL 3.8.5, PROJ 9.5.1; sf_use_s2() is TRUE
```

```
library(tmap)
```

Load in the HOLC (redlining), EJScreen, and SWITRS (crashes) data.

```
# Load/read in the necessary data
ejscreen <- st_read("data/ejscreen/EJSCREEN_2023_BG_StatePct_with_AS_CNMI_GU_VI.gdb")
```

```
Reading layer `EJSCREEN_StatePctiles_with_AS_CNMI_GU_VI' from data source
  `/Users/oMac/EDS223-Homework-2/data/ejscreen/EJSCREEN_2023_BG_StatePct_with_AS_CNMI_GU_VI.g
  using driver `OpenFileGDB'
Simple feature collection with 243021 features and 223 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -19951910 ymin: -1617130 xmax: 16259830 ymax: 11554350
Projected CRS: WGS 84 / Pseudo-Mercator
```

```
crashes <- read_csv("data/ejscreen/SWITRS_LA/Crashes.csv")
```

```
Rows: 251100 Columns: 80
-- Column specification --------------------------------------------------------
Delimiter: ","
chr  (45): COLLISION_TIME, OFFICER_ID, REPORTING_DISTRICT, CHP_BEAT_TYPE, CI...
dbl  (33): CASE_ID, ACCIDENT_YEAR, JURIS, DAY_OF_WEEK, CHP_SHIFT, POPULATION...
date  (2): PROC_DATE, COLLISION_DATE

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
holc <- st_read("data/ejscreen/mapping-inequality/mapping-inequality-los-angeles.json")
```

```
Reading layer `mapping-inequality-los-angeles' from data source
  `/Users/oMac/EDS223-Homework-2/data/ejscreen/mapping-inequality/mapping-
inequality-los-angeles.json'
  using driver `GeoJSON'
Simple feature collection with 417 features and 14 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -118.6104 ymin: 33.70563 xmax: -117.7028 ymax: 34.30388
Geodetic CRS:  WGS 84
```

Part 1: Legacy of redlining in current environmental (in)justice

(a). Create a map of historical redlining neighborhoods

Filter to only Los Angeles County data (our area of interest).

```
# Filter to data of interest
la <- ejscreen |>
      dplyr::filter(CNTY_NAME == "Los Angeles County")
    # Only entries in LA County
la_shape <- st_as_sf(la)
holc_shape <- st_as_sf(holc)
# Save data as spatial for plotting
```

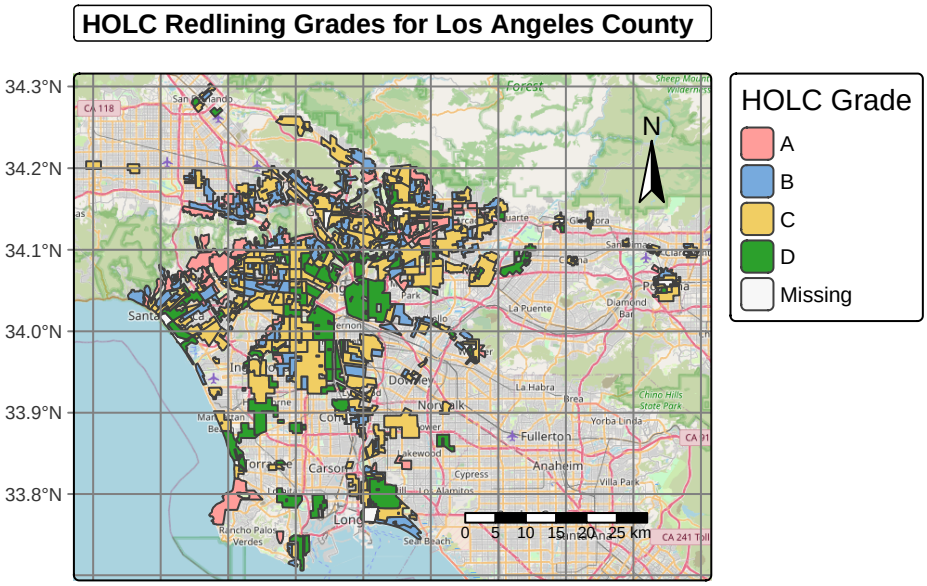
Map Los Angeles County's historical redlining "grades".

```
# Make the map!
tmap_mode("plot")
```

```
i tmap modes "plot" - "view"
i toggle with `tmap::ttm()`
```

```
la_map <- tm_shape(holc_shape) +
    tm_polygons(fill = "grade", fill.legend = tm_legend(title = "HOLC Grade")) +
    tm_title(text = "HOLC Redlining Grades for Los Angeles County", size = 1.1, fontface = "
    tm_basemap("OpenStreetMap") +
    # Choose a base map
    tm_graticules() +
    tm_scalebar(position = tm_pos_in("right", "bottom")) +
    tm_compass(position = tm_pos_in("right", "top"))
    # Add compass, scale bar, scale lines
la_map
```

```
[plot mode] fit legend/component: Some legend items or map compoments do not
fit well, and are therefore rescaled.
i Set the tmap option `component.autoscale = FALSE` to disable rescaling.
```



(b). Table summarizing the percentage of census block groups that fall within each HOLC grade

First, spatially combine the EJScreen and HOLC (redlining) data so that we can look at the percentage of census block groups in Los Angeles County that fall into each of the different HOLC grades.

```
# First we need to combine the EJScreen and HOLC data spatially
# We need to make sure the CRS match!
if(st_crs(ejscreen) == st_crs(holc)){
  print("CRS match")
}else{
  print("CRS do not match")
}
```

```
[1] "CRS do not match"
```

```r
# The CRS do not match, so we need to transform one of the datasets so that they match
holc_transformed <- st_transform(holc, crs = st_crs(ejscreen))

# Check that the CRS now match
if(st_crs(ejscreen) == st_crs(holc_transformed)){
  print("CRS match")
}else{
  print("CRS do not match")
}
```

```
[1] "CRS match"
```

```r
# The CRS now match, so we can now join the datasets
# We only care about the LA data from EJScreen
ej_and_holc <- st_join(la, holc_transformed, join = st_intersects)

# Now that the data is joined, we can drop geometries
ej_and_holc <- st_drop_geometry(ej_and_holc)
```

With the HOLC and EJScreen datasets joined, the next step is to summarize the percentage of census block groups that fall within each HOLC grade.

```r
# Make summary table
summ_table <- ej_and_holc |>
    count(grade) |>
    mutate(percentage = round(n / sum(n) * 100, 1))
    # Calculate the percentages in each grade
print(summ_table)
```

```
  grade    n percentage
1     A  449        5.0
2     B 1239       13.8
3     C 3058       34.0
4     D 1346       15.0
5  <NA> 2896       32.2
```

(c). Visualizations summarizing current conditions within HOLC grades for % low income, percentile for Particulate Matter 2.5, and percentile for low life expectancy

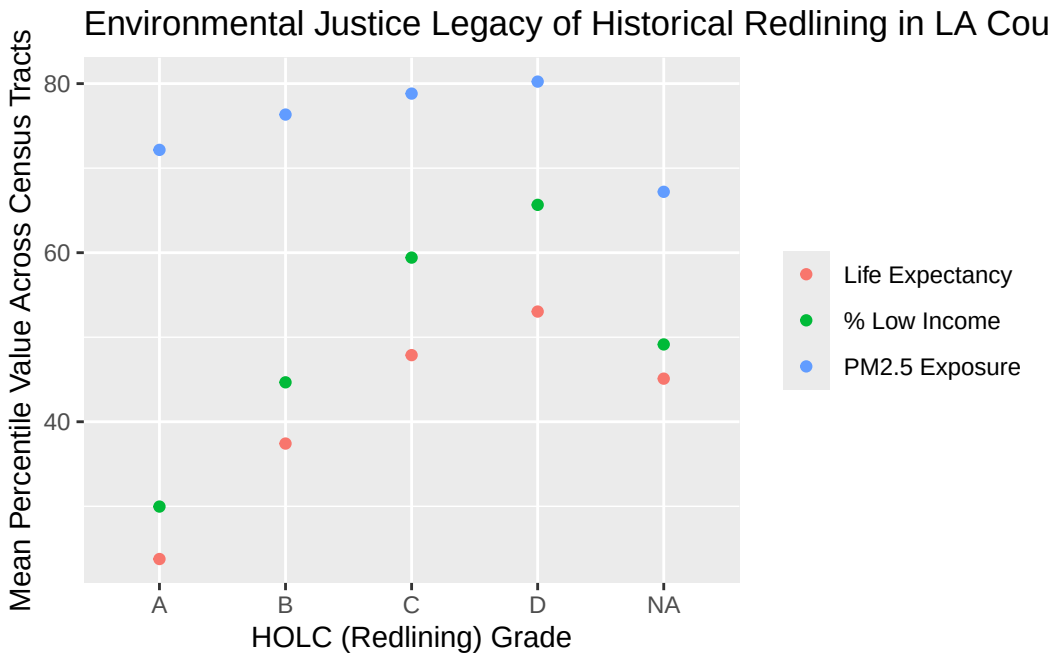
With the EJScreen and HOLC (redlining) data spatially joined, we can look at various environmental justice variables (PM2.5 exposure, low income areas, and life expectancy) across the different HOLC grades.

```r
# First take the means of these variables, grouped by HOLC grade
holc_vars_summ <- ej_and_holc |>
    group_by(grade) |>
    summarize(mean(P_LOWINCPCT, na.rm = TRUE), mean(P_PM25, na.rm = TRUE), mean(P_LIFEEXPPCT
```

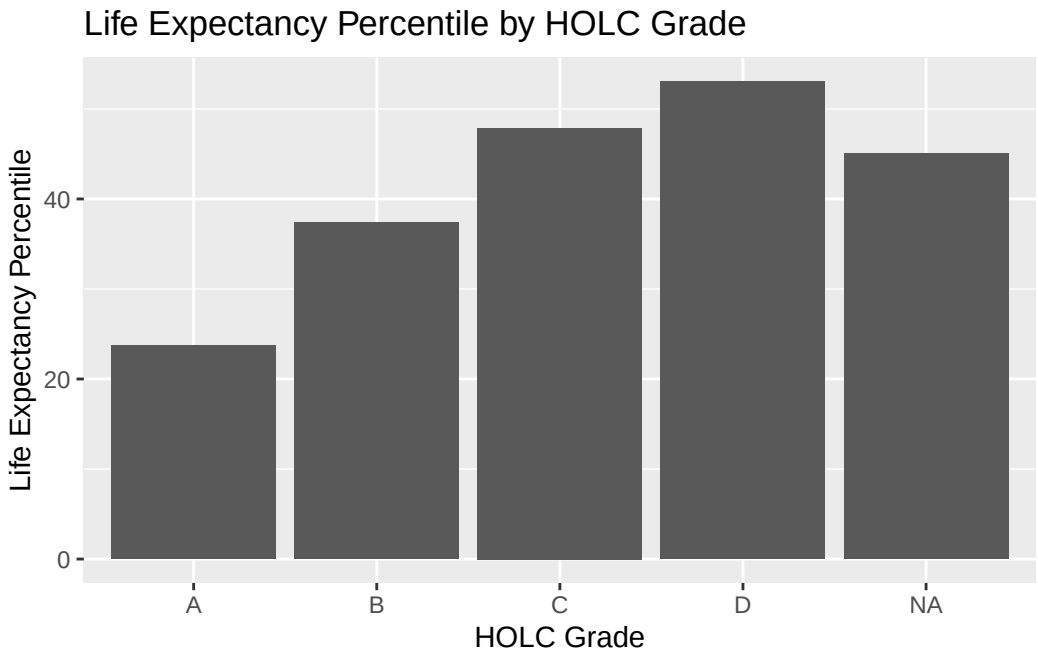
Using ggplot, create a scatterplot as well as barplots of the mean percentile for each of these three variables (PM2.5 exposure, low income areas, and life expectancy) within each HOLC grade.

```r
# Visualize as a scatterplot!
# Need to pivot longer first so we have one row per grade/variable combo
holcvars_long <- holc_vars_summ |>
  pivot_longer(
    cols = c(`mean(P_LOWINCPCT, na.rm = TRUE)`, `mean(P_PM25, na.rm = TRUE)`, `mean(P_LIFEEX
    names_to = "variable",
    values_to = "percentile"
  )

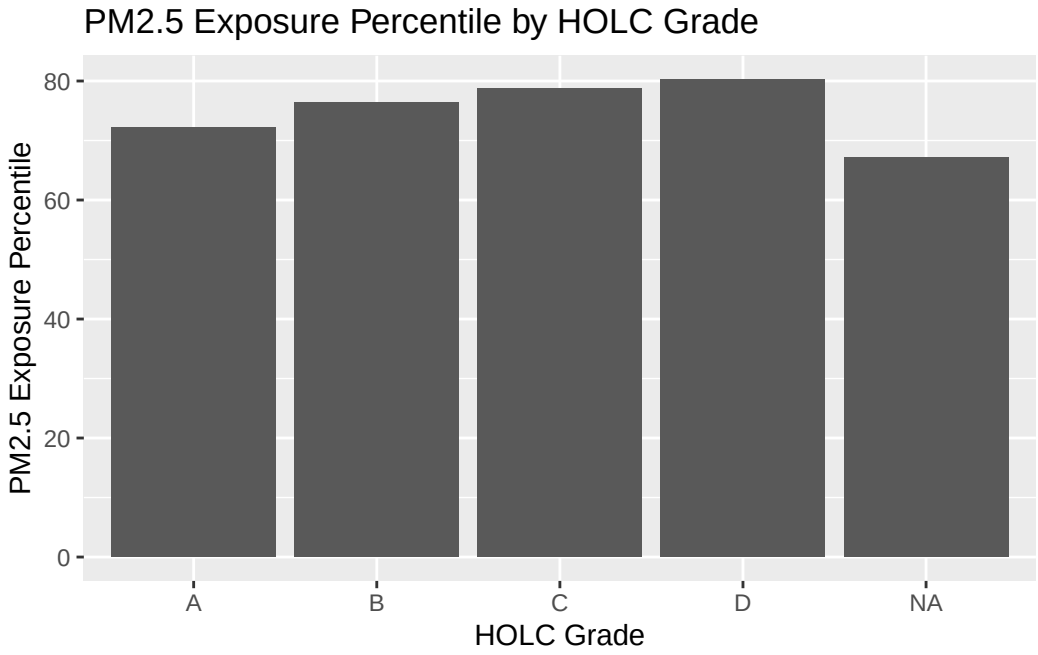
# Plot a scatterplot using ggplot
ggplot(holcvars_long, aes(
        x = grade,
        y = percentile,
        color = variable)) +
    geom_point() +
    scale_color_discrete(
    # Color by variable
        labels = c(
            `mean(P_LOWINCPCT, na.rm = TRUE)` = "% Low Income",
            `mean(P_PM25, na.rm = TRUE)` = "PM2.5 Exposure",
            `mean(P_LIFEEXPPCT, na.rm = TRUE)` = "Life Expectancy")
    ) +
    labs(x = "HOLC (Redlining) Grade", y = "Mean Percentile Value Across Census Tracts", col
```



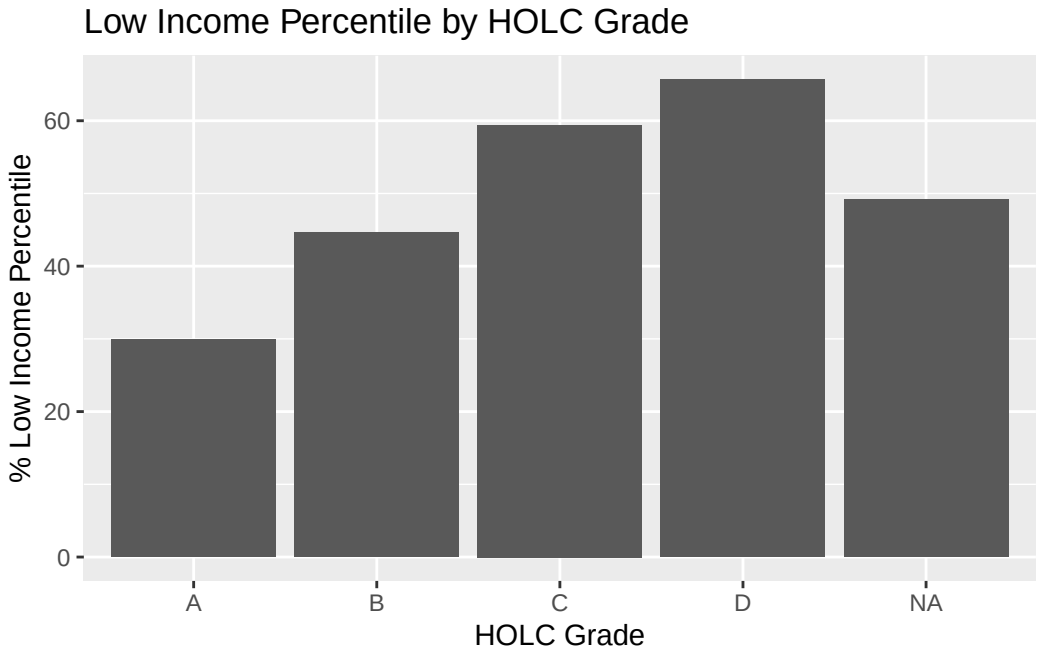

```
# We can also visualize individual variables with barplots
# Life expectancy
holc_lifeexp <- ej_and_holc |>
    group_by(grade) |>
    summarize(mean(P_LIFEEXPPCT, na.rm = TRUE))
life_exp_plot <- ggplot(holc_lifeexp, aes(x = grade, y = `mean(P_LIFEEXPPCT, na.rm = TRUE)`))
    geom_bar(stat = "identity") +
    labs(x = "HOLC Grade", y = "Life Expectancy Percentile", title = "Life Expectancy Percent
life_exp_plot
```




```
# PM2.5
holc_pm25 <- ej_and_holc |>
    group_by(grade) |>
    summarize(mean(P_PM25, na.rm = TRUE))
pm25_plot <- ggplot(holc_pm25, aes(x = grade, y = `mean(P_PM25, na.rm = TRUE)`)) +
    geom_bar(stat = "identity") +
    labs(x = "HOLC Grade", y = "PM2.5 Exposure Percentile", title = "PM2.5 Exposure Percenti
pm25_plot
```




```
# Low income percentile
holc_lowinc <- ej_and_holc |>
    group_by(grade) |>
    summarize(mean(P_LOWINCPCT, na.rm = TRUE))
lowinc_plot <- ggplot(holc_lowinc, aes(x = grade, y = `mean(P_LOWINCPCT, na.rm = TRUE)`)) +
    geom_bar(stat = "identity") +
    labs(x = "HOLC Grade", y = "% Low Income Percentile", title = "Low Income Percentile by H
lowinc_plot
```



(d). Reflection paragraph

Based on these visualizations, it appears that the average (mean) percentile across all three variables (percent low income, percentile for particulate matter 2.5, and percentile for low life expectancy) increases as the grade decreases (i.e. areas with grade "D" will tend to be in a higher percentile for low income %, PM2.5, and low life expectancy than areas with an "A" grade). This trend is observed similarly across all three variables. This trend is likely a result of historical redlining, as these visualizations show that "redlined" areas now display a lower average life expectancy, lower average income, and more PM2.5 air pollution. In this way, the discriminatory legacy of historical redlining persists today through these variables.

Part 2: Legacy of redlining in pedstrian and bike accidents

(a). Figure summarizing the percent of observations within redlined neighborhoods within each HOLC grade

Before any visualization can happen, we need to join the HOLC (redlining "grades") data with the SWITRS (crashes) dataset.

```
# We need to join the HOLC data with the crashes data
# "Crashes" is not spatial data, so we do not need to worry about CRS matching right now

# To join "crashes" to "holc", we need to combine the latitude/longitude columns in "crashes"
# and convert it to a spatial object
# We also need to drop NAs!
crash_clean <- crashes |>
    filter(!is.na(POINT_X), !is.na(POINT_Y))
    # Filter to only rows where the coordinates do not contain NA values
    # POINT_X and POINT_Y have less NA values than the lat/long columns
latlong <- st_as_sf(crash_clean, coords = c("POINT_X", "POINT_Y"), crs = st_crs(holc))
# Create spatial data by combining the coordinates
# And set CRS to that of the HOLC data
```

```
# Now we check that the CRS match
if(st_crs(latlong) == st_crs(holc)){
  print("CRS match")
}else{
  print("CRS do not match")
}
```

```
[1] "CRS match"
```

```
# Finally, we can now join the HOLC and crashes datasets
holc <- st_make_valid(holc)
holc_and_crash <- st_join(latlong, holc, join = st_within)
```

After joining the data, we can look at the percentage of total crashes that fall within each HOLC grade.

```
# We can now drop geometry
holc_and_crash <- st_drop_geometry(holc_and_crash)

# First, lets look at the number of crashes within each grade
crash_summ <- holc_and_crash |>
    count(grade) |>
    mutate(percentage = round(n / sum(n) * 100, 1))
```

Based on this table, it appears that the majority of crashes occur in areas given an “NA” grade - therefore, it will be helpful to drop NA values before continuing.

```r
# We may need to drop NAs to make the data cleaner
crash_summ_no_na <- holc_and_crash |>
    filter(!is.na(grade)) |>
    # Remove NA values
    count(grade) |>
    mutate(percentage = round(n / sum(n) * 100, 1))

crash_summ_no_na
```

```
# A tibble: 4 x 3
  grade     n percentage
  <chr> <int>      <dbl>
1 A      2706        3.1
2 B     11398       13.2
3 C     45770       53.1
4 D     26295       30.5
```

After removing NAs, it seems that the majority (53%) of observations in the "crashes" dataset occur in areas given the HOLC grade "C". Looking at specific crash types (bike incidents, pedestrian incidents, and fatal crashes), this pattern holds true.

```r
# Looking specifically at bike crashes
bike_crashes <- holc_and_crash |>
    filter(BICYCLE_ACCIDENT == "Y")
    # Filter to only bike incidents
    # (i.e. where the value in the "BICYCLE_ACCIDENT" column is "Y")
bike_summ <- bike_crashes |>
    filter(!is.na(grade)) |>
    # Remove NA values
    count(grade) |>
    mutate(percentage = round(n / sum(n) * 100, 1))
    # What percentage falls in each grade?
bike_summ
```

```
# A tibble: 4 x 3
  grade     n percentage
  <chr> <int>      <dbl>
1 A       173        3.1
2 B       612       11.1
3 C      3077       55.7
4 D      1660       30.1
```

```r
# Looking specifically at pedestrian crashes
ped_crashes <- holc_and_crash |>
    filter(PEDESTRIAN_ACCIDENT == "Y")
   # Filter to only pedestrian incidents
ped_summ <- ped_crashes |>
    filter(!is.na(grade)) |>
    # Remove NA values
    count(grade) |>
    mutate(percentage = round(n / sum(n) * 100, 1))
    # What percentage falls in each grade?
ped_summ
```

```
# A tibble: 4 x 3
  grade     n percentage
  <chr> <int>      <dbl>
1 A       187        1.8
2 B      1171       11.4
3 C      5980       58.4
4 D      2901       28.3
```

```r
# Looking specifically at fatalities
fatal_crashes <- holc_and_crash |>
    filter(NUMBER_KILLED >= 1.00)
    # Filter to only fatal incidents
    # (i.e. where "NUMBER_KILLED" is greater than or equal to 1)
fatal_summ <- fatal_crashes |>
    filter(!is.na(grade)) |>
    # Remove NA values
    count(grade) |>
    mutate(percentage = round(n / sum(n) * 100, 1))
    # What percentage falls in each grade?
fatal_summ
```

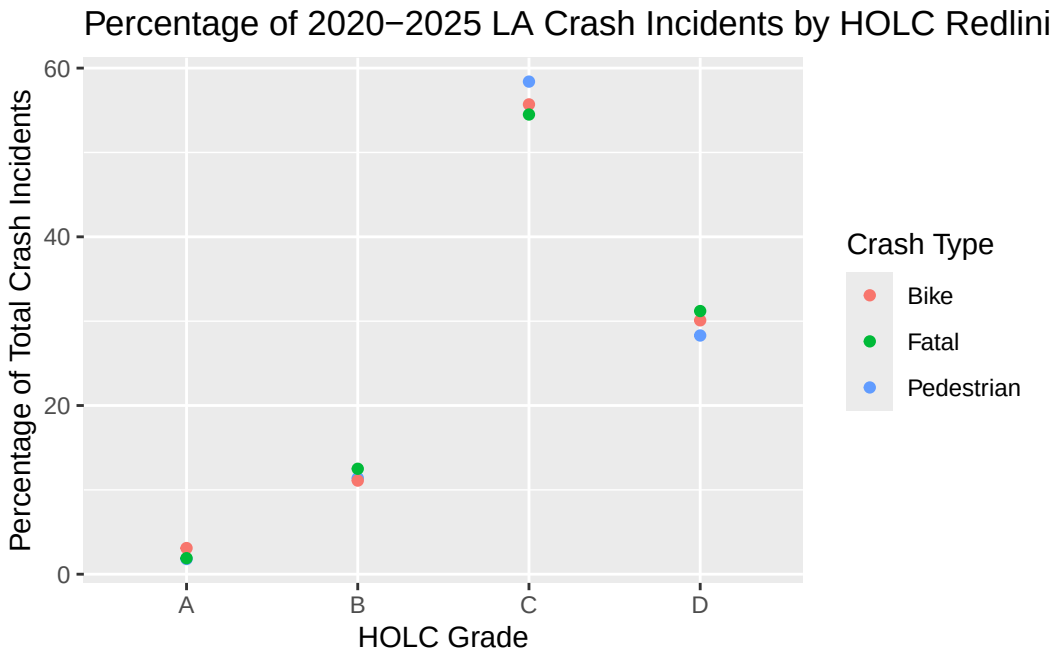
```
# A tibble: 4 x 3
  grade     n percentage
  <chr> <int>      <dbl>
1 A        25        1.9
2 B       167       12.5
3 C       730       54.5
4 D       418       31.2
```

Across all three of these categories (bike crashes, pedestrian crashes, and fatal crashes), the majority of incidents (54%-58%) occur in areas with a HOLC grade of "C", followed by "D" (28%-31%), then "B" (11%-13%) and finally "A" (1%-3%). Therefore, the number of incidents mostly increases with decreasing grade, with the exception of "C" graded areas, which have a higher number of incidents than "D".

Finally, the summarized data can be visualized as a scatterplot using ggplot.

```
# Now that the data is clear, we can use ggplot to visualize it
# first we need to add crash type identifiers to the tables
# And then combine the summary tables into one table
bike_summ <- bike_summ |>
    mutate("crash_type" = "Bike")
    # Add a "crash_type" column so that we can differentiate between types once the data is j
ped_summ <- ped_summ |>
    mutate("crash_type" = "Pedestrian")
fatal_summ <- fatal_summ |>
    mutate("crash_type" = "Fatal")
bike_ped_fatal <- bind_rows(ped_summ, bike_summ, fatal_summ)
# Create a data frame of all 3 crash types within HOLC grades

# Finally we can plot the data!
crash_plot <- ggplot(bike_ped_fatal, aes(x = grade, y = percentage, color = crash_type)) +
    geom_point() +
    labs(x = "HOLC Grade", y = "Percentage of Total Crash Incidents", title = "Percentage of
crash_plot
```


Percentage of 2020−2025 LA Crash Incidents by HOLC Redlini
Percentage of Total Crash Incidents
60
40
20
0
A
B
C
D
HOLC Grade
Crash Type
Bike
Fatal
Pedestrian